Extension: Minimum Price Contract for Wheat Output

Progress Update — AgEnRes Deliverable 5.4

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Outline

1. Recap: Input Price Risk
2. Extension: Wheat Minimum Price Contract
3. Contract Payoff Structure
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Previous Work: Fertilizer Max-Price Contract

- European **call option** on CAN fertilizer price
- Farmer pays premium ρ; if spot price exceeds strike K, excess is covered
- Effective price paid: $\min(p_x^s, K) + \rho$

Key finding:

- Input price risk has a **limited impact** on total profit variability
- Profit variability mainly driven by **output prices and yields**

$\rightarrow$ Need to also address **output price risk**

New: Wheat Minimum Price Contract (Put Option)

Idea

Add a **price floor** on wheat revenue using a European put option — symmetric to the fertilizer price ceiling.

- Farmer pays premium ρ_w; if wheat spot price falls below strike K_w, the shortfall is covered
- Effective price received: $\max(p_w^s, K_w) - \rho_w$
- Addresses **downside output price risk** (revenue protection)

Additionally: Evaluate a **combined contract** (fertilizer ceiling + wheat floor) to address both input and output risk simultaneously.

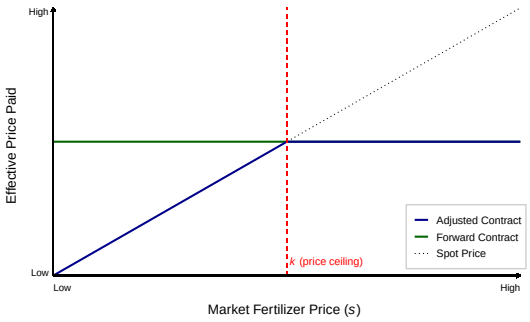
Contract Design: Full Comparison

	Fertilizer call	**Wheat put**	**Combined**
Type	Call option	Put option	Call + Put
Risk hedged	Input cost ↑	Revenue ↓	Both
Effective price	$\min(p_x^s, K_f)$	$\max(p_w^s, K_w)$	Both
Premium	ρ_f (€/kg)	ρ_w (€/t)	$\rho_f + \rho_w$
Premium (per ha)	$\rho_f \times \text{CAN}$	$\rho_w \times y$	sum
Strike quantiles	50th–90th	10th–50th	Diagonal pairing
Maturity	6 months	12 months	Both

Diagonal pairing: fertilizer strike i is paired with wheat strike i (rank-matched).

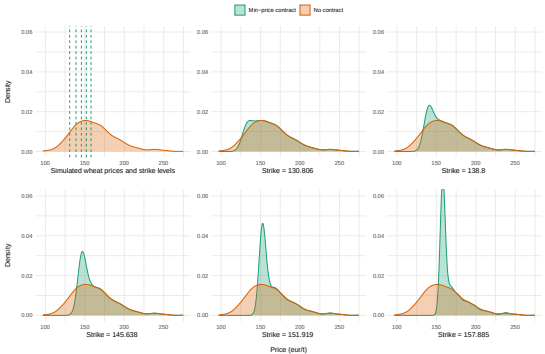
Payoff Diagrams

Fertilizer: Price Ceiling (Call)



Effective price **paid**: $\min(p_x^s, K_f)$

Wheat: Price Floor (Put)



Effective price **received**: $\max(p_w^s, K_w)$

Contract Pricing: Fertilizer Call vs. Wheat Put

Fertilizer call option

Quantile	Strike (€/kg)	Premium (€/kg)
0.5	0.355	0.011
0.6	0.361	0.008
0.7	0.368	0.005
0.8	0.376	0.004
0.9	0.388	0.002

Higher strike $\rightarrow$ lower premium

Wheat put option

Quantile	Strike (€/t)	Premium (€/t)
0.1	130.81	0.82
0.2	138.80	1.95
0.3	145.64	3.60
0.4	151.92	5.75
0.5	157.89	8.33

Higher strike $\rightarrow$ higher premium

Wheat put premiums are larger in absolute terms but small relative to wheat prices (0.5–5.3% of strike).

17 Profit Scenarios

Label	Contract	Profit formula
nc	No contract	$p_w^s \cdot y - p_x^s \cdot \text{CAN}$
fc	Forward (fertilizer)	$p_w^s \cdot y - p_x^f \cdot \text{CAN}$
ac_1...5	Fert. ceiling only	$p_w^s \cdot y - [\min(p_x^s, K_f) + \rho_f] \cdot \text{CAN}$
wc_1...5	Wheat floor only	$[\max(p_w^s, K_w) - \rho_w] \cdot y - p_x^s \cdot \text{CAN}$
both_1...5	Combined (diagonal)	$[\max(p_w^s, K_w) - \rho_w] \cdot y - [\min(p_x^s, K_f) + \rho_f] \cdot \text{CAN}$

- All evaluated under **1,000 simulated price paths** (copula-linked)
- EUT and CPT applied to all 17 columns automatically
- Broad framing: ref = $E[\pi^{fc}]$ | Narrow framing: ref = 0

Profit Summary Statistics (€/ha)

Scenario	Mean	Median	SD	Min	Max
No contract (nc)	1140.01	1096.79	311.50	272.42	2296.22
Forward (fc)	1137.58	1099.02	319.41	255.29	2385.43
ac_1 (fert. ceiling)	1139.88	1097.88	316.35	264.28	2380.36
ac_5	1139.99	1097.83	313.30	270.94	2362.58
wc_1 (wheat floor)	1140.31	1092.97	304.24	**382.24**	2287.53
wc_5	**1141.91**	1117.20	**265.93**	420.25	2207.99
both_1 (combined)	1140.18	1095.56	309.02	374.10	2372.44
both_5	1141.88	1116.64	267.64	418.77	2282.19

- Wheat floor raises minimum profit substantially (+370 → +420 €/ha at wc_5)

EUT: Broad Framing — Certainty Equivalents (€/ha)

r	nc	fc	ac (fert. ceiling)			wc (wheat floor)			both (combined)		
			1	3	5	1	3	5	1	3	5
10^{-5}	1139.5	1137.1	1139.4	1140.0	1139.5	1139.9	1140.6	**1141.6**	1139.7	1141.1	1141.5
10^{-4}	1135.2	1132.5	1134.9	1135.6	1135.1	1135.7	1136.9	**1138.4**	1135.4	1137.3	1138.3
10^{-3}	1091.5	1086.6	1089.8	1091.0	1090.9	1094.0	1099.8	**1106.6**	1092.4	1099.4	1106.1
10^{-2}	897.4	882.5	889.7	892.7	894.6	908.9	935.3	**965.1**	901.5	931.1	962.8

- Wheat floor (wc_5) yields **highest CE** across all risk aversion levels
- At $r = 10^{-3}$: wc_5 = **1107** vs. nc = 1092 vs. ac_3 = 1091 €/ha (+15 €/ha)
- Combined contract performs nearly identically to wheat-only floor
- Advantage **grows with risk aversion**: at $r = 10^{-2}$, gap reaches 68 €/ha

CPT: Broad Framing — Prospect Values

	nc	fc	ac					wc					both				
			1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
Standard	−1.90	−1.93	−1.91	−1.90	−1.89	−1.90	−1.89	−1.87	−1.85	−1.79	−1.75	−1.69	−1.87	−1.85	−1.79	−1.75	−1.69
$\alpha = 0.5$	−4.54	−4.64	−4.53	−4.52	−4.49	−4.52	−4.51	−4.36	−4.28	−4.13	−4.03	−3.91	−4.35	−4.25	−4.08	−4.01	−3.87
$\lambda = 2$	−3.23	−3.28	−3.25	−3.25	−3.23	−3.24	−3.23	−3.19	−3.16	−3.07	−3.01	−2.93	−3.20	−3.17	−3.07	−3.02	−2.93

- All prospect values remain **negative** — but wheat floor **improves** them
- wc_5 and both_5: $V = -1.69$ vs. nc $= -1.90$ (11% less negative)
- Improvement is consistent across CPT parametrizations
- Still negative $\rightarrow$ CPT farmer would not voluntarily adopt, but gap is narrowing

EUT: Narrow Framing — Certainty Equivalents (€/ha)

r	fc	ac (fert. ceiling)			wc (wheat floor)			both (combined)		
		1	3	5	1	3	5	1	3	5
0	**2.43**	−0.13	0.51	−0.03	0.21	0.94	2.17	0.08	1.45	2.14
10^{-4}	**2.29**	−0.19	0.47	−0.04	−0.09	−0.60	−1.65	−0.24	−0.04	−1.61
10^{-3}	**1.81**	−0.39	0.35	−0.09	−1.07	−5.67	−14.17	−1.29	−4.95	−13.95

- Forward contract **dominates** under narrow framing (consistent with previous finding)
- At risk neutrality ($r = 0$): wc_5 CE = 2.17, close to fc = 2.43
- But at higher r: wheat floor CE **turns deeply negative** (high payoff variance)
- Combined contracts inherit this high-variance problem

CPT: Narrow Framing — Prospect Values

	fc	ac					wc					both				
		1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
Standard	−0.52	−0.65	−0.65	−0.63	−0.81	−0.78	−1.34	−1.44	−1.44	−1.39	−1.30	−0.85	−1.16	−1.32	−1.39	−1.31
$\lambda = 0.5$	**0.75**	0.49	0.44	0.39	0.24	0.16	0.24	0.42	0.62	0.80	**0.98**	0.57	0.57	0.67	0.79	0.97

- Wheat floor contracts are **most negative** under standard CPT ($V \approx -1.3$ to -1.4)
- Large downside (premium paid) triggers strong **loss aversion**
- Exception: loss-seeking ($\lambda = 0.5$) — wc_5 achieves highest $V = 0.98$
- Combined contracts slightly better than wheat-only but still worse than forward

Summary of Results

	Broad framing	Narrow framing
EUT	**Wheat floor / Combined** (highest CE; grows with r)	Forward contract (dominates; options too volatile)
CPT	Wheat floor best among options (still $V < 0$)	Forward least negative (wheat put too loss-averse)

Key insights:

- Under **broad framing + EUT**, addressing output price risk provides **substantial welfare gains** (up to 68 €/ha at high risk aversion)
- Under **CPT**, wheat floor **narrows the gap** but prospect values remain negative
- Under **narrow framing**, wheat floor's high payoff variance makes it **unattractive**
- Framing effect is **even stronger** for output contracts than for input contracts

Next Steps

1. **Sensitivity analysis:** Systematic CPT parameter sweeps for wc and both contracts
2. **Non-diagonal pairing:** Explore all 25 fertilizer×wheat strike combinations
3. **Endogenous reference points:** Kőszegi–Rabin formulation
4. **Write up** for Deliverable 5.4 report

Thank you!

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